Titanic Dataset – Exploratory Data Analysis (EDA)

By: ANAND UMRANIA

🔰 Importing Libraries

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import missingno as msno
import warnings

warnings.filterwarnings('ignore')
sns.set(style='whitegrid')
```

📂 Loading the Datasets

In [4]:
```
train = pd.read_csv('Dataset/train.csv')
test = pd.read_csv('Dataset/test.csv')
gender_sub = pd.read_csv('Dataset/gender_submission.csv')
```

🔍 Previewing the Datasets

In [8]:
```
# Display the first 5 rows
train.head()
```

Out[8]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [10]:

```
# Display basic information about the dataset
train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [12]:
```
# Checking for missing values in training data
train.isnull().sum()
```

Out[12]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

🧩 Combining Train and Test Datasets for Comprehensive EDA

In [15]:
```
# Add a 'source' column to distinguish between train and test
train['source'] = 'train'
test['source'] = 'test'
test['Survived'] = np.nan

# Concatenate both datasets
combined = pd.concat([train, test], ignore_index=True)
combined.head()
```

Out[15]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	source
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	train
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	train
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	train
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S	train
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S	train

🧼 Missing Value Analysis & Treatment

🔍 Check Missing Values (in % and counts)

In [19]:

```
# Calculate total and percentage of missing values in each column
missing_count = combined.isnull().sum()
missing_percent = (missing_count / combined.shape[0]) * 100

missing_df = pd.DataFrame({
    'Missing Values': missing_count,
    'Percentage': missing_percent.round(2)
}).sort_values(by='Percentage', ascending=False)

missing_df
```

Out[19]:

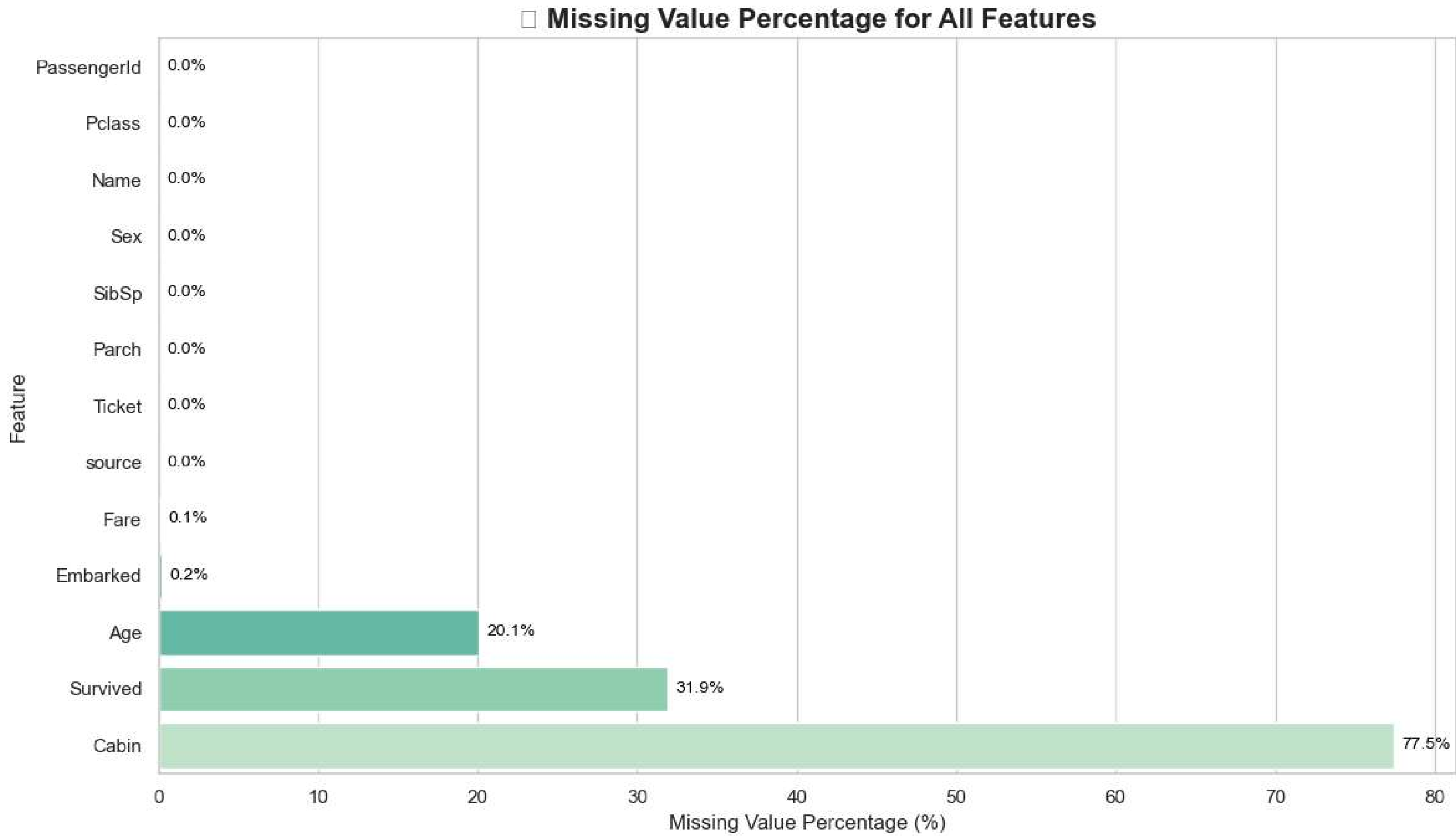
	Missing Values	Percentage
Cabin	1014	77.46
Survived	418	31.93
Age	263	20.09
Embarked	2	0.15
Fare	1	0.08
PassengerId	0	0.00
Pclass	0	0.00
Name	0	0.00
Sex	0	0.00
SibSp	0	0.00
Parch	0	0.00
Ticket	0	0.00
source	0	0.00

📊 Visualize Missing Values

In [22]:

```
#  Get all features and calculate missing %
all_features = combined.columns
missing = combined.isnull().sum()
missing_percent = (missing / len(combined)) * 100

missing_percent = missing_percent[all_features].sort_values(ascending=True)

# Plot with all features (including 0%)
plt.figure(figsize=(12, 7))
sns.set_theme(style="whitegrid")
colors = sns.color_palette("mako", len(missing_percent))
```

```
sns.barplot(
    x=missing_percent.values,
    y=missing_percent.index,
    palette=colors
)

for index, value in enumerate(missing_percent):
    plt.text(value + 0.5, index, f'{value:.1f}%', va='center', fontsize=10, color='black')

plt.title('📊 Missing Value Percentage for All Features', fontsize=16, fontweight='bold')
plt.xlabel('Missing Value Percentage (%)', fontsize=12)
plt.ylabel('Feature', fontsize=12)
plt.tight_layout()
plt.show()
```



🔧 Handle Missing Values

In [25]:
```
# Fill Embarked with most frequent value
combined['Embarked'].fillna(combined['Embarked'].mode()[0], inplace=True)

# Fill Fare with median
combined['Fare'].fillna(combined['Fare'].median(), inplace=True)

# Fill Age with median
combined['Age'].fillna(combined['Age'].median(), inplace=True)
```

```
# For Cabin, extract only deck letter if present, else mark as 'M' for missing
combined['Deck'] = combined['Cabin'].apply(lambda x: x[0] if pd.notnull(x) else 'M')
```

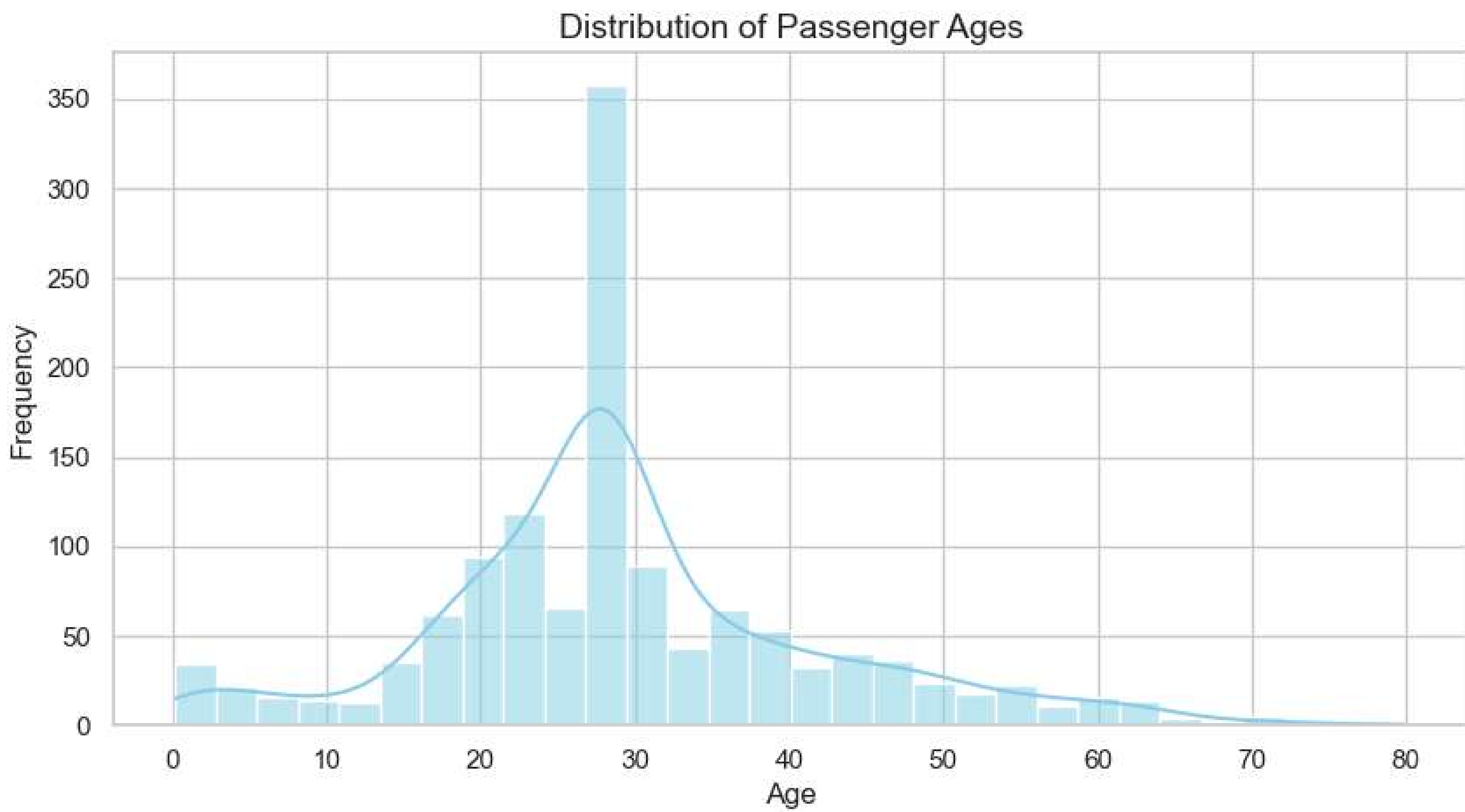
In [27]:
```
combined.isnull().sum()
```

Out[27]:
```
PassengerId       0
Survived        418
Pclass            0
Name              0
Sex               0
Age               0
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin          1014
Embarked          0
source            0
Deck              0
dtype: int64
```

📈 Univariate Analysis (One Feature at a Time)

🔢 Numerical Features

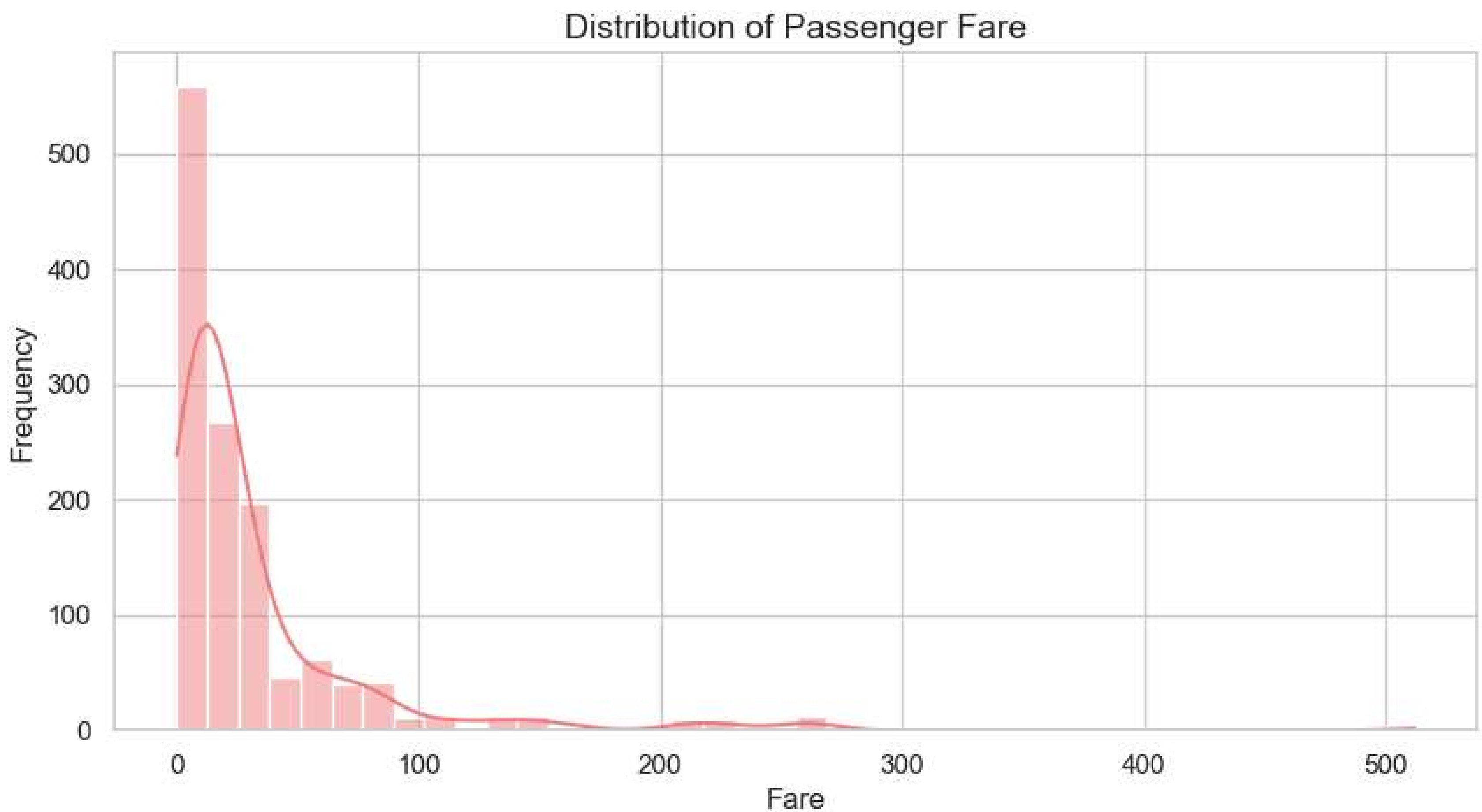
In [31]:
```
plt.figure(figsize=(10, 5))
sns.histplot(data=combined, x='Age', bins=30, kde=True, color='skyblue')
plt.title('Distribution of Passenger Ages', fontsize=14)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



📌 Observations: - Most passengers are between 20 and 40 years old. - A sharp drop-off occurs beyond 60+. - Distribution is slightly right-skewed.

📊 Fare Distribution

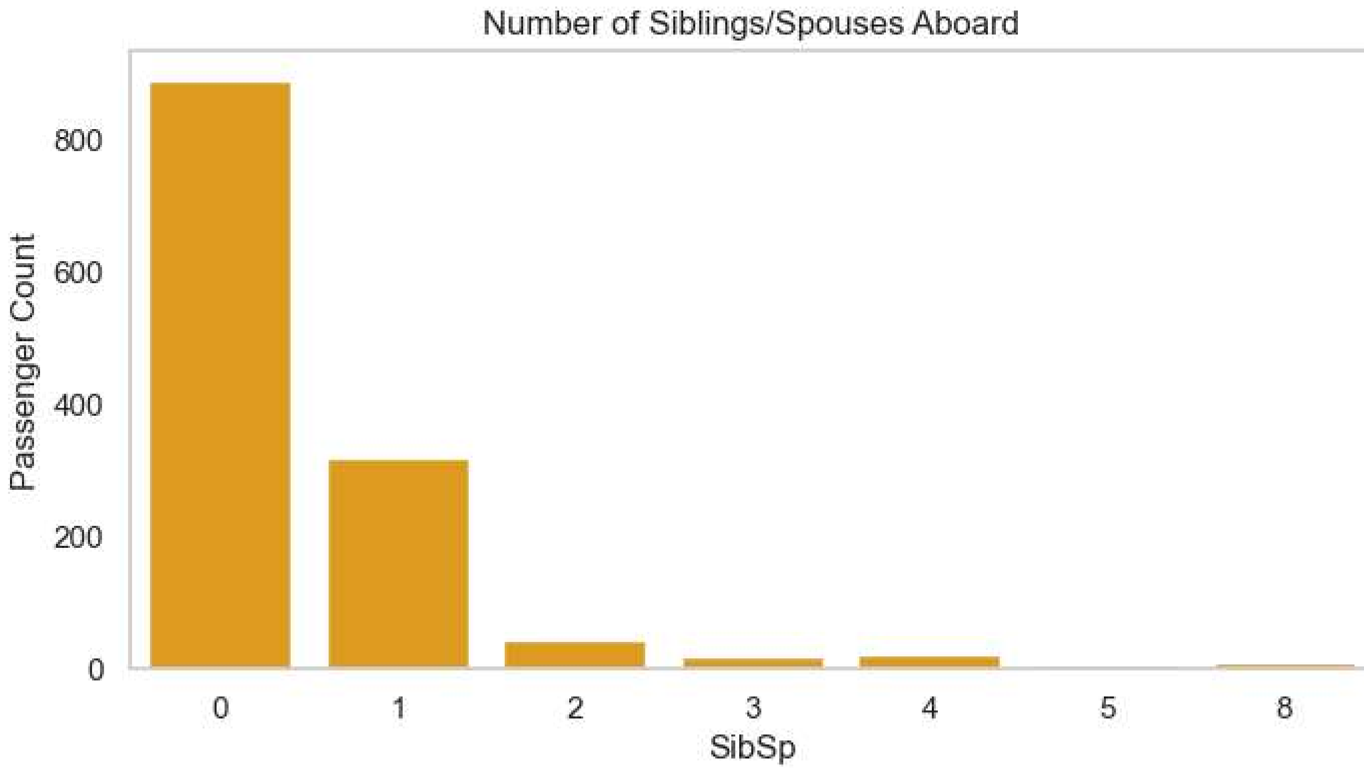
In [34]:
```
plt.figure(figsize=(10, 5))
sns.histplot(data=combined, x='Fare', bins=40, kde=True, color='lightcoral')
plt.title('Distribution of Passenger Fare', fontsize=14)
plt.xlabel('Fare')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



📌 Observations: - Most passengers are between 20 and 40 years old. - A sharp drop-off occurs beyond 60+. - Distribution is slightly right-skewed.

📊 SibSp (Siblings/Spouses aboard)

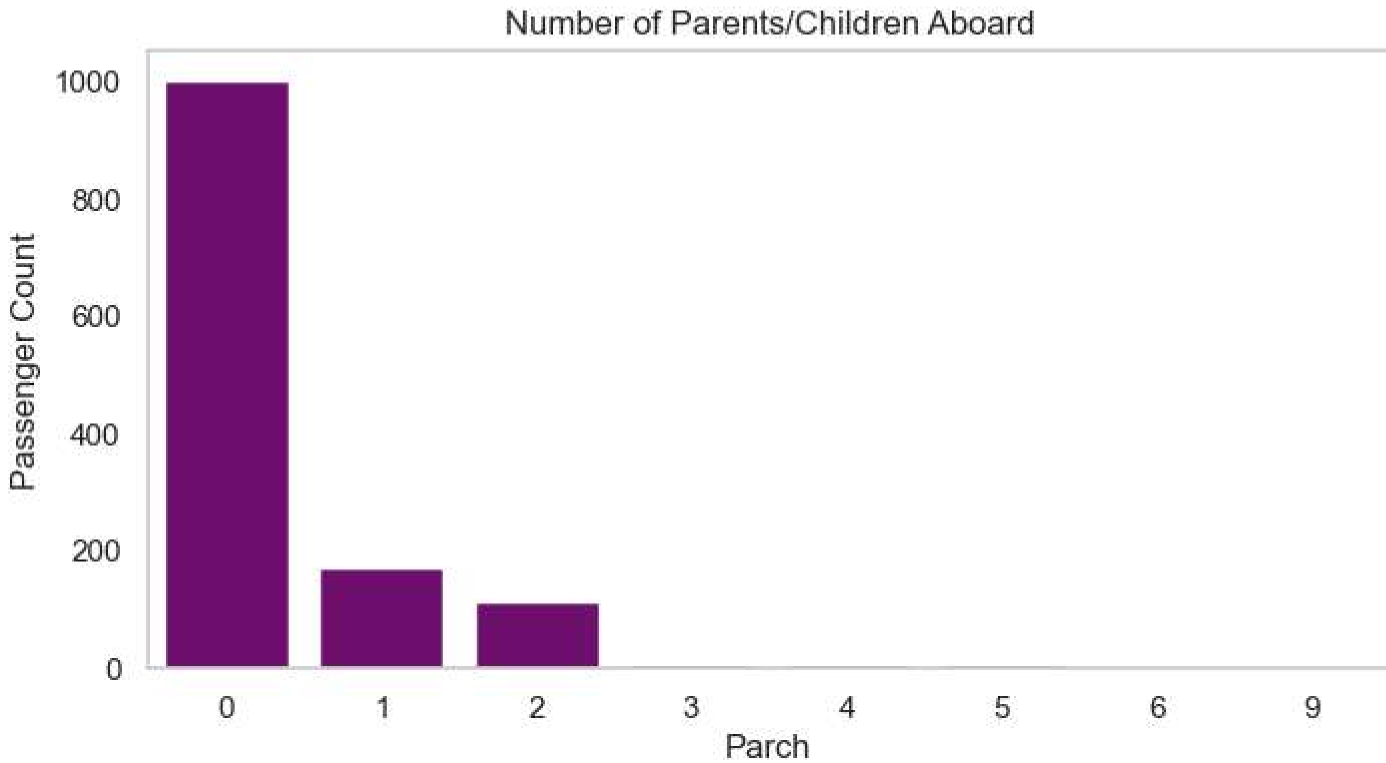
In [37]:
```
plt.figure(figsize=(8, 4))
sns.countplot(data=combined, x='SibSp', color='orange')
plt.title('Number of Siblings/Spouses Aboard')
plt.xlabel('SibSp')
plt.ylabel('Passenger Count')
plt.grid(axis='y')
plt.show()
```



📌 Observations: - Most passengers had no siblings or spouses aboard. - A few traveled with 1 or 2 family members.

📊 Parch (Parents/Children aboard)

In [40]:
```
plt.figure(figsize=(8, 4))
sns.countplot(data=combined, x='Parch', color='purple')
plt.title('Number of Parents/Children Aboard')
plt.xlabel('Parch')
plt.ylabel('Passenger Count')
plt.grid(axis='y')
plt.show()
```

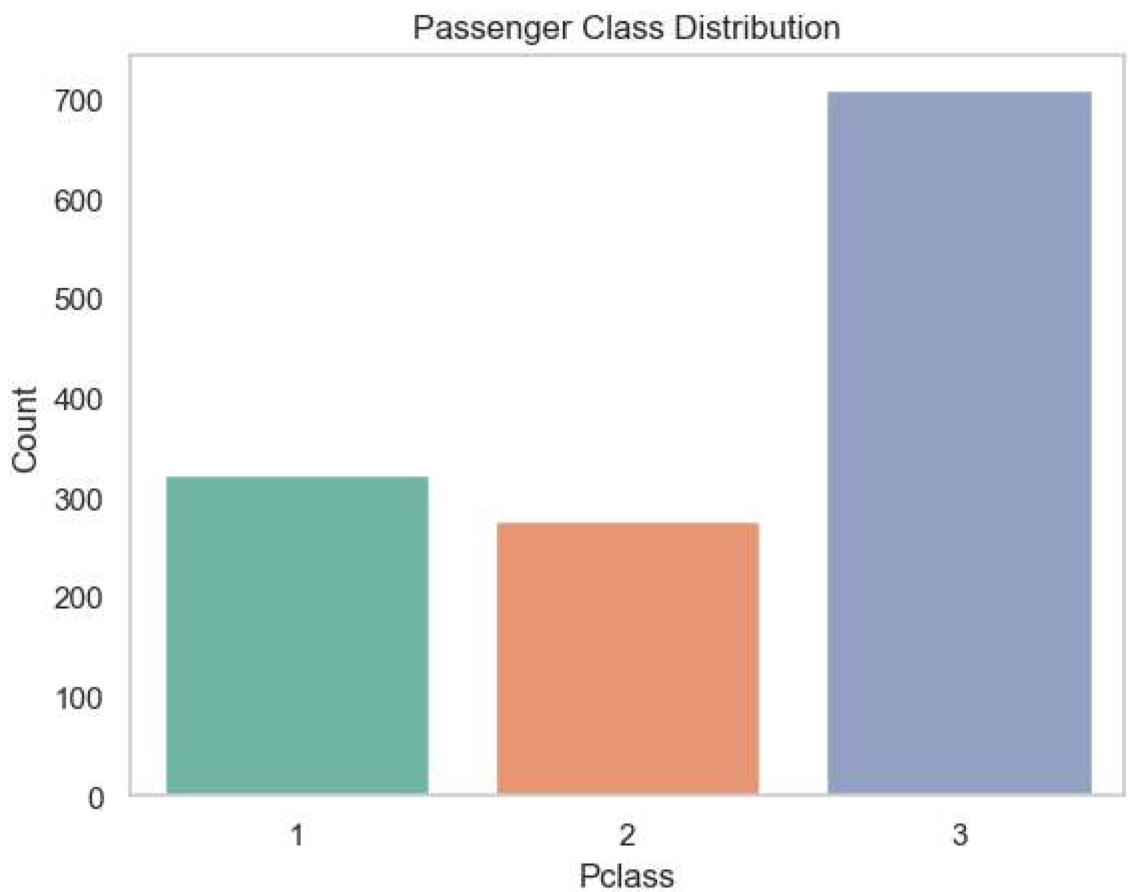


📌 Observations: - Most passengers were not traveling with parents/children. - A few large families are present.

🔠 Categorical Features

📊 Passenger Class Distribution

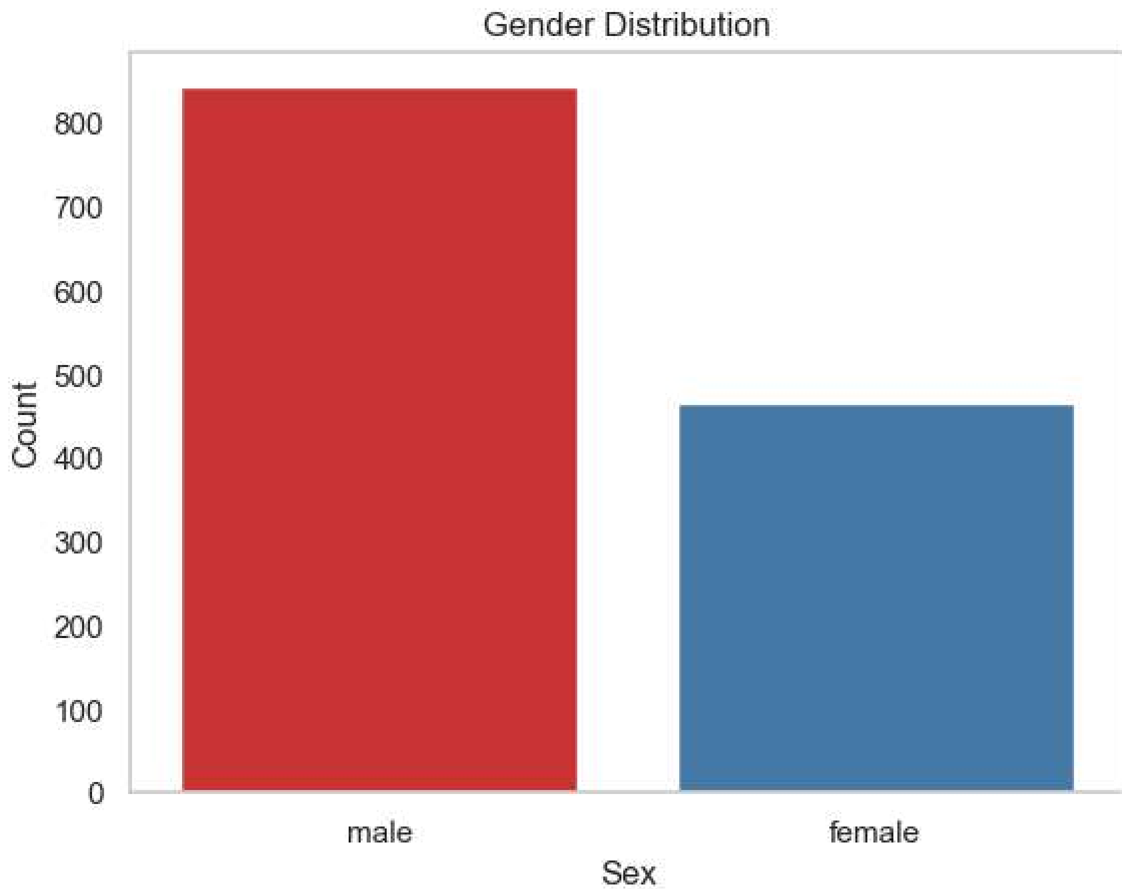
In [44]:
```
sns.countplot(data=combined, x='Pclass', palette='Set2')
plt.title('Passenger Class Distribution')
plt.xlabel('Pclass')
plt.ylabel('Count')
plt.grid(axis='y')
plt.show()
```



📌 Observations: - Most passengers were in 3rd class. - Very few were in 1st class.

📊 6.2.2 Gender Distribution

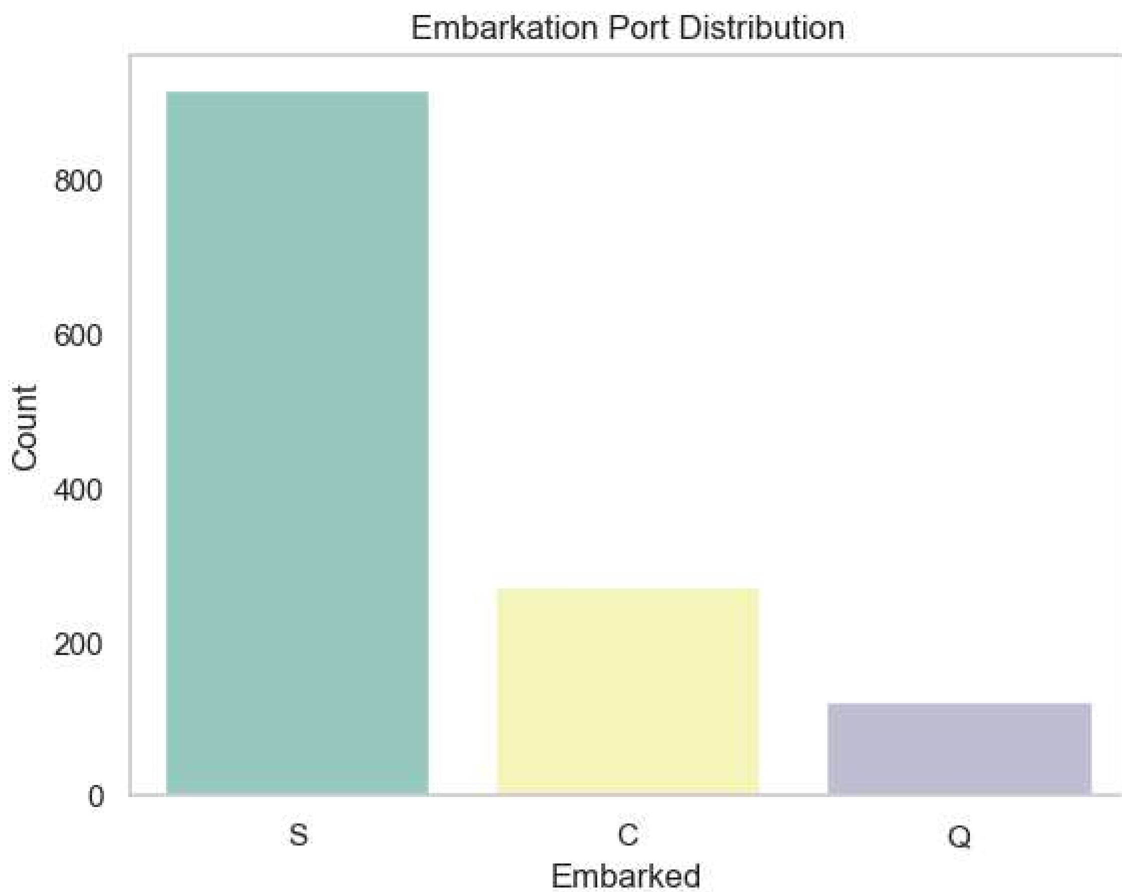
In [47]:
```
sns.countplot(data=combined, x='Sex', palette='Set1')
plt.title('Gender Distribution')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.grid(axis='y')
plt.show()
```



📌 Observations: - More male passengers than female.

📊 Embarked Port Distribution

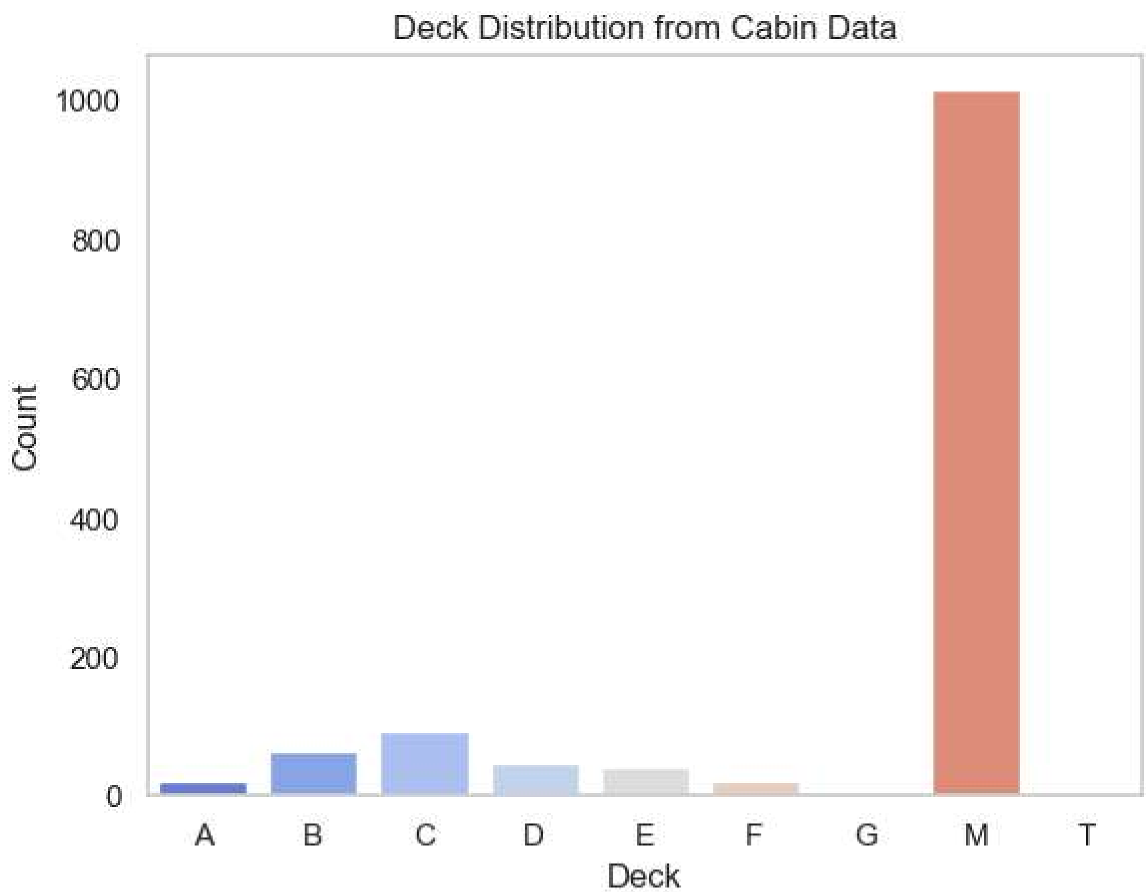
In [50]:
```
sns.countplot(data=combined, x='Embarked', palette='Set3')
plt.title('Embarkation Port Distribution')
plt.xlabel('Embarked')
plt.ylabel('Count')
plt.grid(axis='y')
plt.show()
```



📌 Observations: - Majority of passengers boarded at port 'S' (Southampton). - 'C' (Cherbourg) and 'Q' (Queenstown) had fewer passengers.

📊 Deck Distribution (from Cabin)

In [53]:
```
sns.countplot(data=combined, x='Deck', order=sorted(combined['Deck'].unique()), palette='coolwarm')
plt.title('Deck Distribution from Cabin Data')
plt.xlabel('Deck')
plt.ylabel('Count')
plt.grid(axis='y')
plt.show()
```



📌 Observations: - Most deck information is missing (`M`). - Decks B, C, D were the most commonly recorded.

📊 Bivariate Analysis (Survived vs Other Features)

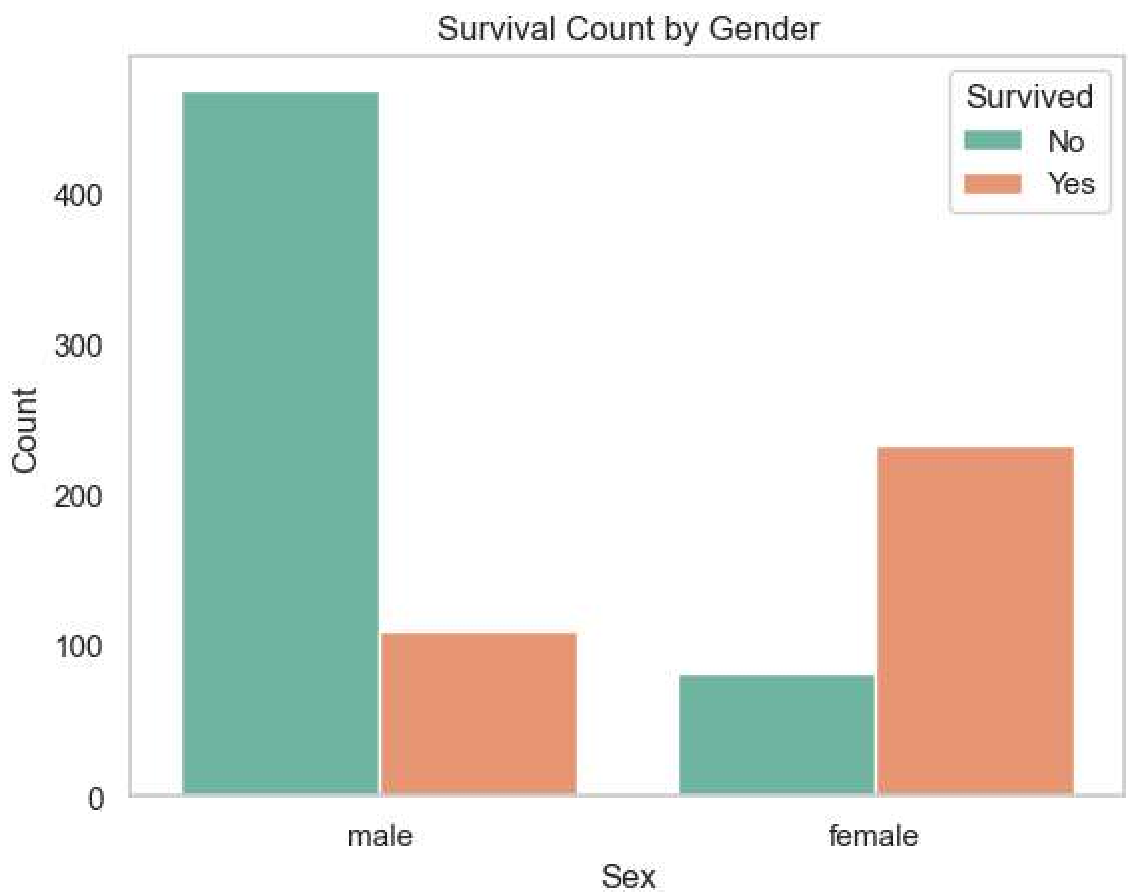
In [56]:
```
# Filter only training data with known survival status
train = combined[combined['source'] == 'train']
```

🧠 Goal: See how features like Sex, Pclass, Age, etc. impact Survived

💙 Survival by Gender

In [59]:
```
sns.countplot(data=train, x='Sex', hue='Survived', palette='Set2')
plt.title('Survival Count by Gender')
```

```
plt.xlabel('Sex')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.grid(axis='y')
plt.show()
```

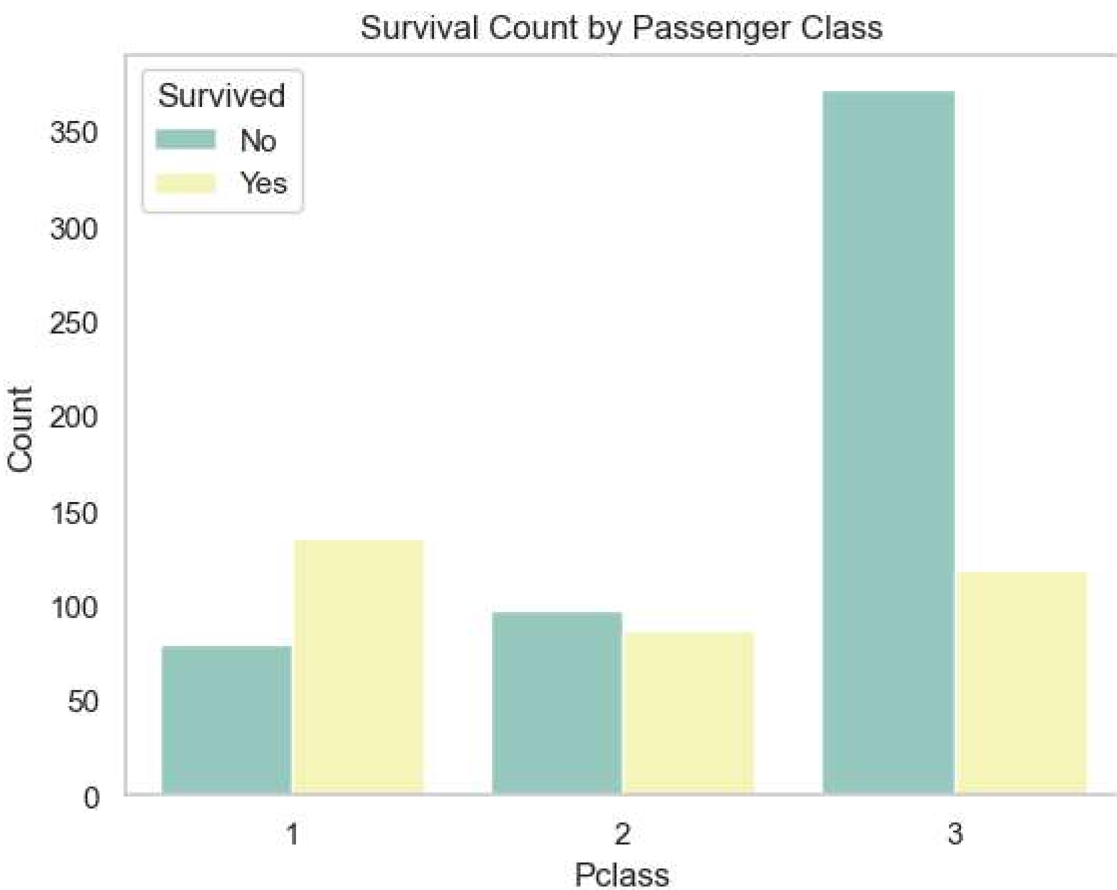


📌 Observation: - Females had a much higher survival rate than males. - Being female strongly increased the chance of survival.

🏷️ Survival by Passenger Class

In [62]:
```
sns.countplot(data=train, x='Pclass', hue='Survived', palette='Set3')
plt.title('Survival Count by Passenger Class')
plt.xlabel('Pclass')
plt.ylabel('Count')
```

```
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.grid(axis='y')
plt.show()
```

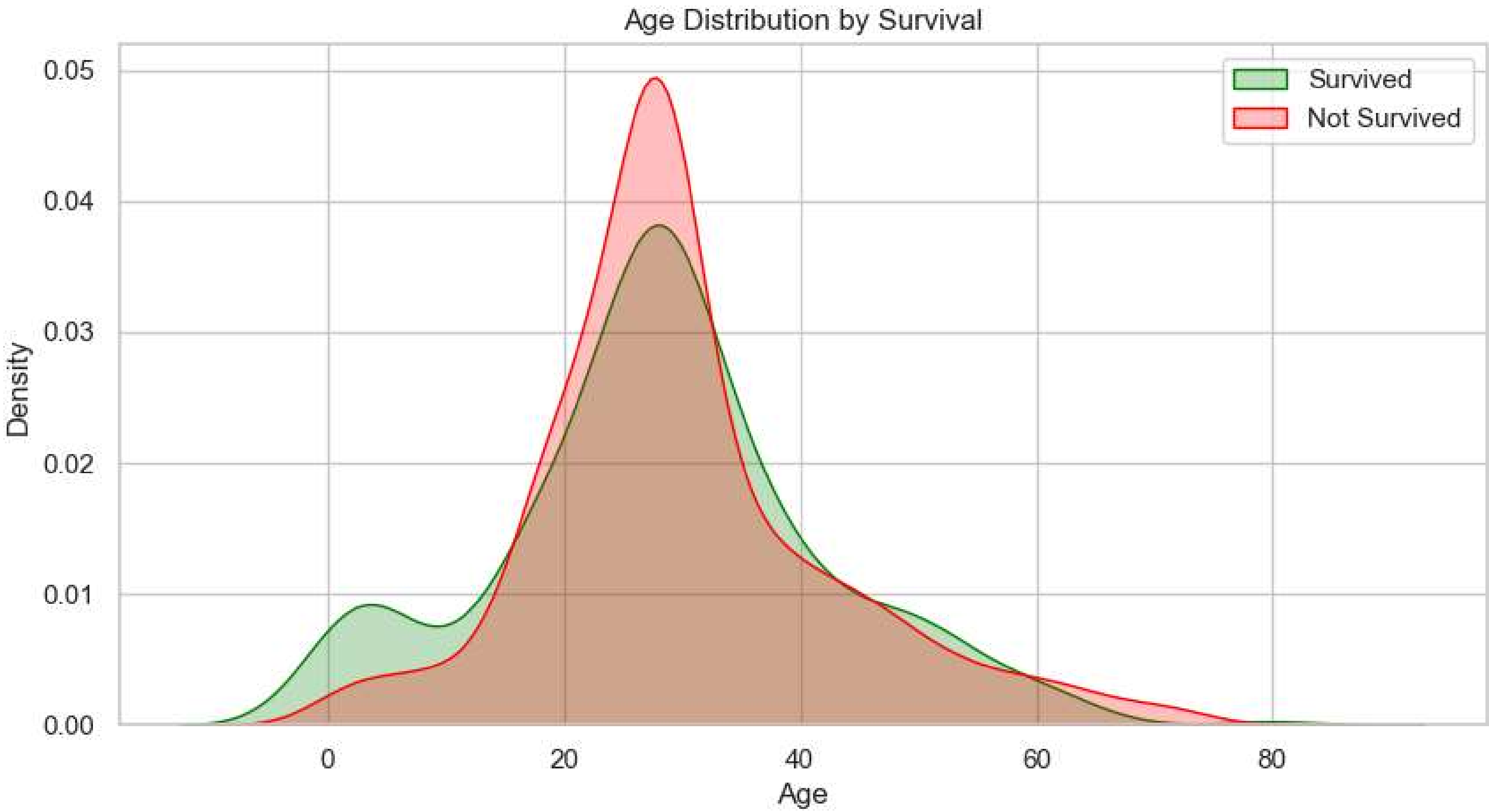


📌 Observation: - 1st class passengers had the highest survival rate. - 3rd class had the highest death count. - Higher class = better survival chance.

🧒 Survival by Age (Using KDE Plots)

In [65]:
```
plt.figure(figsize=(10,5))
sns.kdeplot(data=train[train['Survived'] == 1], x='Age', shade=True, label='Survived', color='green')
sns.kdeplot(data=train[train['Survived'] == 0], x='Age', shade=True, label='Not Survived', color='red')
plt.title('Age Distribution by Survival')
plt.xlabel('Age')
plt.ylabel('Density')
```

```
plt.legend()
plt.grid(True)
plt.show()
```

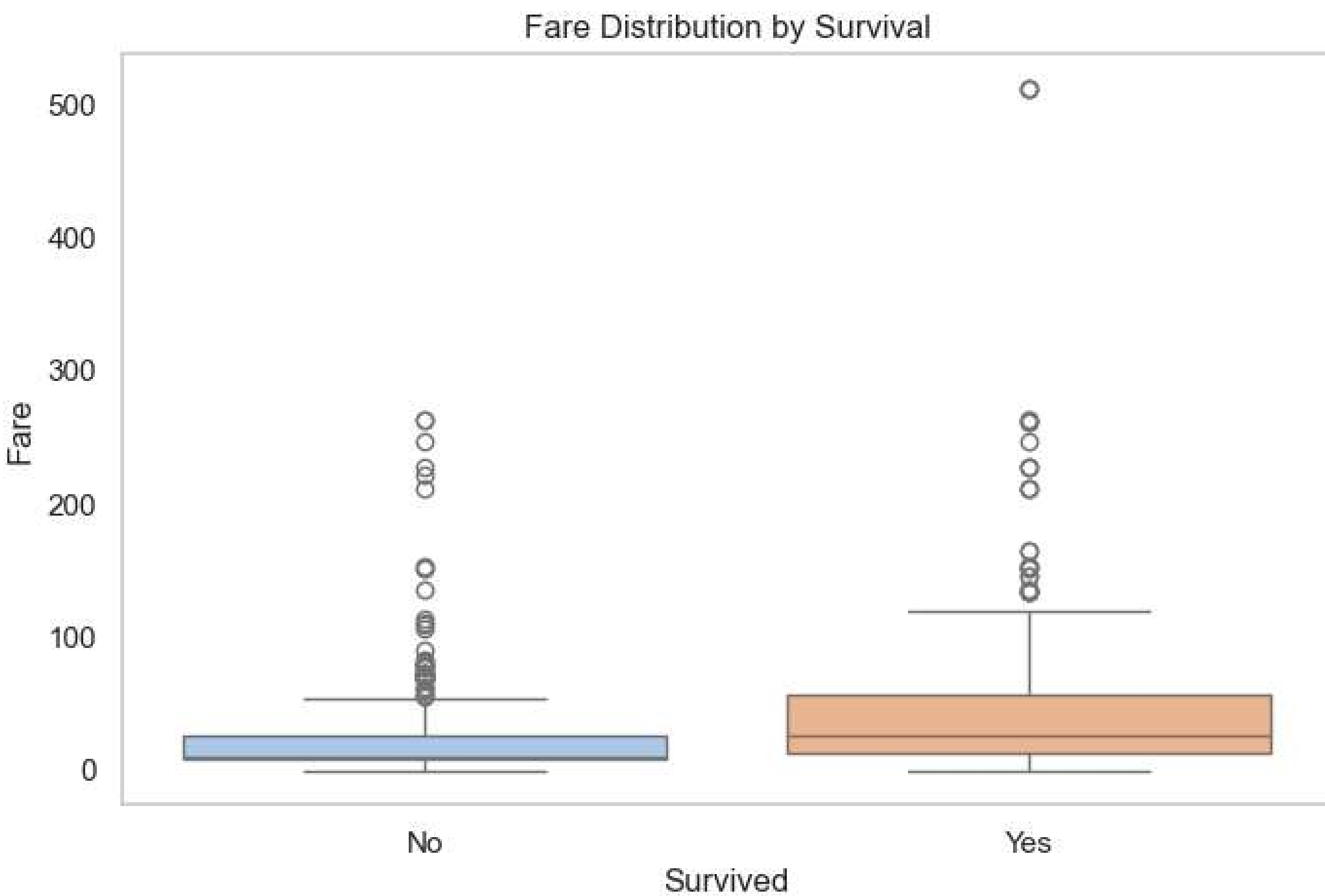


📌 Observation: - Children had slightly better survival. - Older adults (>60) had lower survival chances. - Young adults (20–40) had the highest number of non-survivors.

🎫 Survival by Fare (Box Plot)

In [69]:
```
plt.figure(figsize=(8,5))
sns.boxplot(data=train, x='Survived', y='Fare', palette='pastel')
plt.title('Fare Distribution by Survival')
plt.xlabel('Survived')
plt.ylabel('Fare')
```

```
plt.xticks([0, 1], ['No', 'Yes'])
plt.grid(axis='y')
plt.show()
```

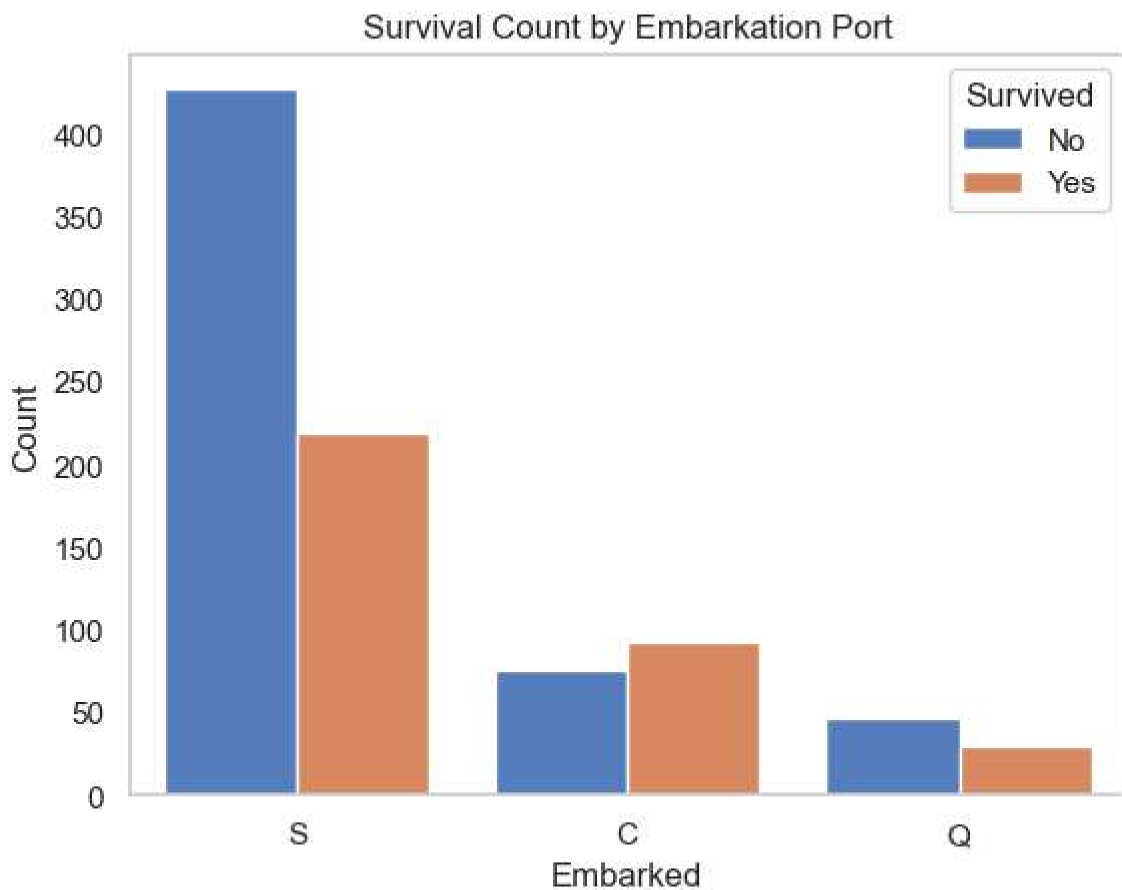


📌 Observation: - Survived passengers had a higher median fare. - Higher fare = likely higher class = better survival chance.

⚓ Survival by Embarked Port

In [72]:
```
sns.countplot(data=train, x='Embarked', hue='Survived', palette='muted')
plt.title('Survival Count by Embarkation Port')
plt.xlabel('Embarked')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
```

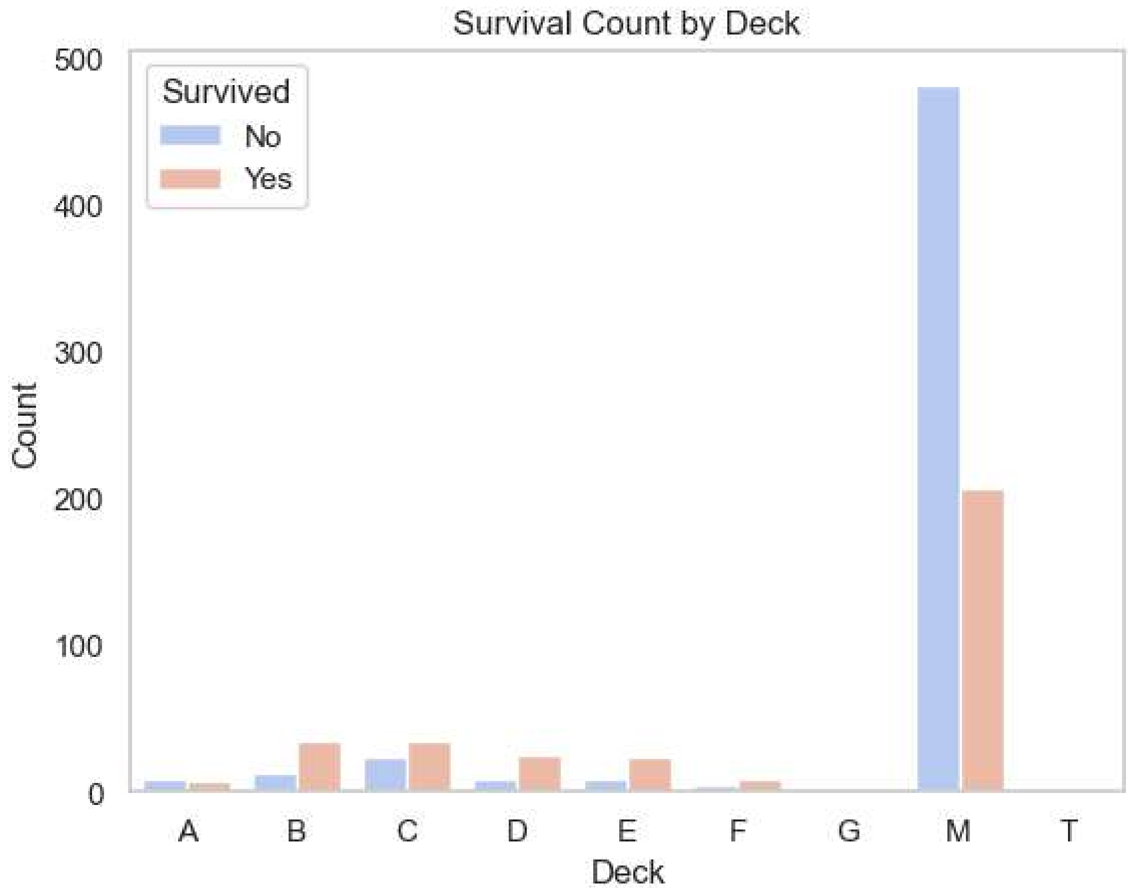
```
plt.grid(axis='y')
plt.show()
```



📌 Observation: - Passengers from port 'C' had higher survival rates. - Those who embarked at 'S' had more non-survivors.

🧱 Survival by Deck

In [75]:
```
sns.countplot(data=train, x='Deck', hue='Survived', palette='coolwarm', order=sorted(train['Deck'].unique()))
plt.title('Survival Count by Deck')
plt.xlabel('Deck')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.grid(axis='y')
plt.show()
```



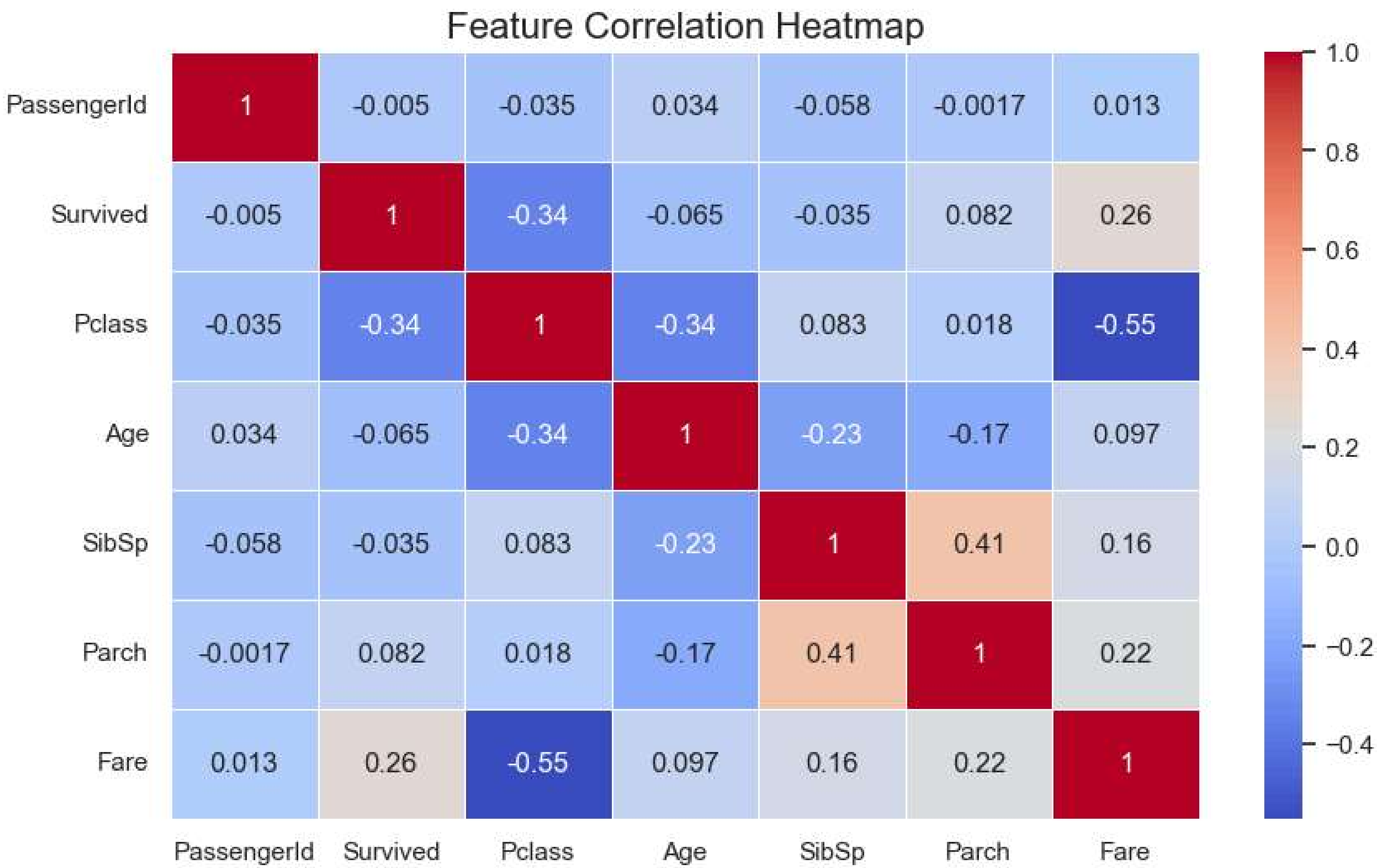
📌 Observation: - Passengers from decks B, D, and E had better survival outcomes. - Decks with more 1st class cabins generally show higher survival.

🔥 Multivariate Analysis (Multiple Variables Together)

📊 Correlation Matrix (Heatmap)

In [79]:
```
# Select only numeric features for correlation
numeric_features = train.select_dtypes(include=['int64', 'float64'])

# Correlation matrix
plt.figure(figsize=(10, 6))
sns.heatmap(numeric_features.corr(), annot=True, cmap='coolwarm', linewidths=0.5)
```

```
plt.title('Feature Correlation Heatmap', fontsize=16)
plt.show()
```

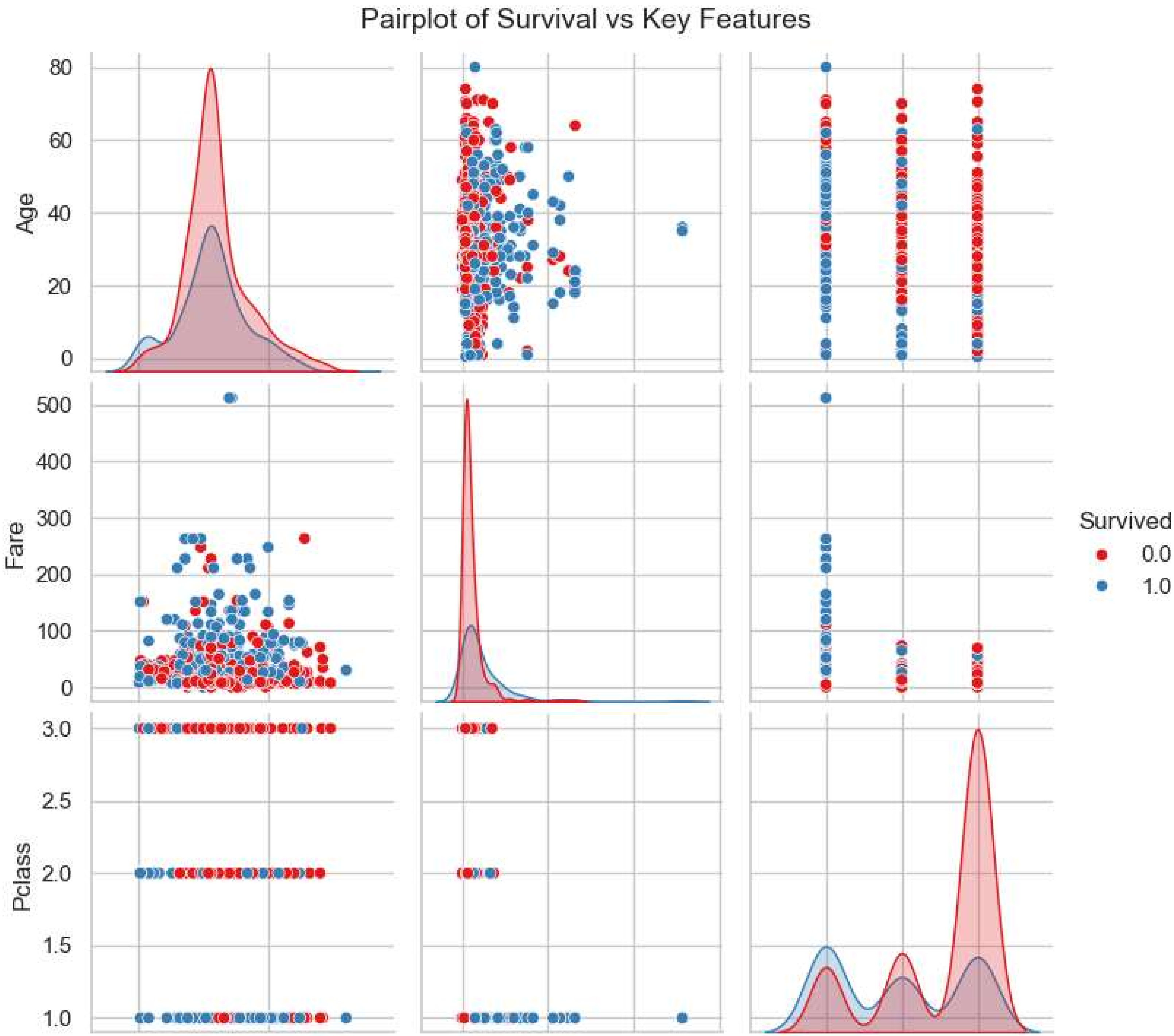


📌 Observations: - Survival is positively correlated with Fare and negatively with Pclass. - SibSp and Parch have slight positive correlation with Survived. - Strong correlation between SibSp and Parch → could combine into FamilySize.

🧩 Pairplot with Key Features

In [82]:
```
sns.pairplot(train[['Survived', 'Age', 'Fare', 'Pclass']], hue='Survived', palette='Set1', diag_kind='kde')
plt.suptitle('Pairplot of Survival vs Key Features', y=1.02, fontsize=14)
```

```
plt.show()
```

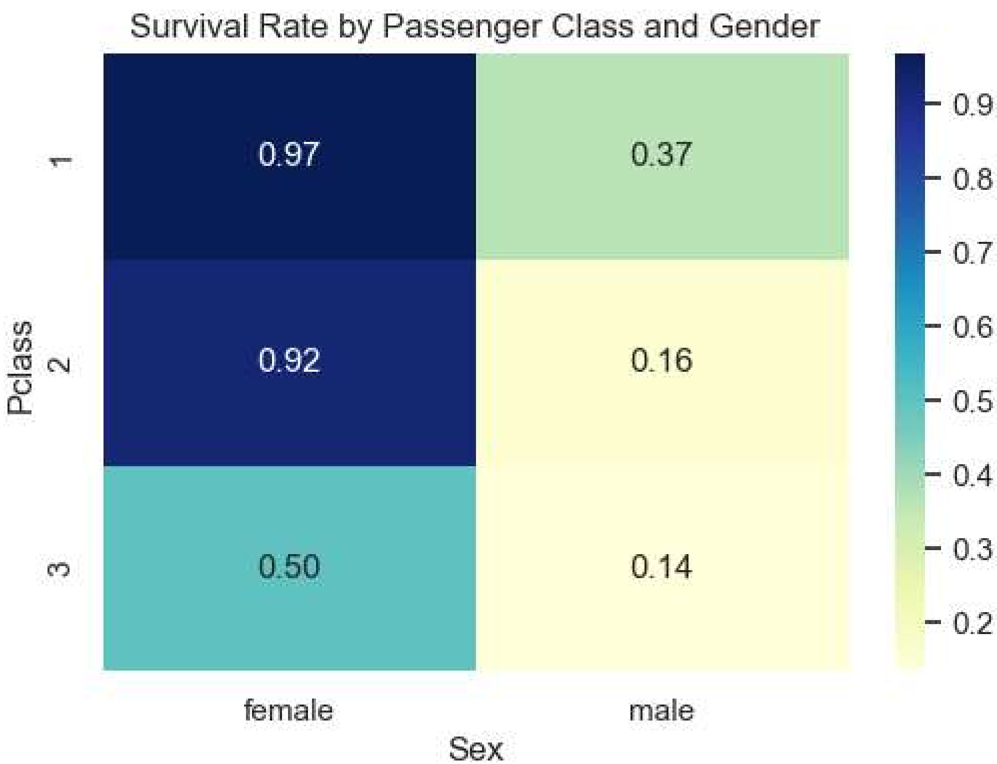

Pairplot of Survival vs Key Features
Age
Fare
Pclass
80
60
40
20
0
500
400
300
200
100
0
3.0
2.5
2.0
1.5
1.0
Survived
0.0
1.0

📌 Observations: - 1st class passengers mostly survived. - Survived passengers tended to pay higher fares. - Survivors are somewhat younger.

🧱 Heatmap of Survival Rates by Pclass & Sex

In [84]:

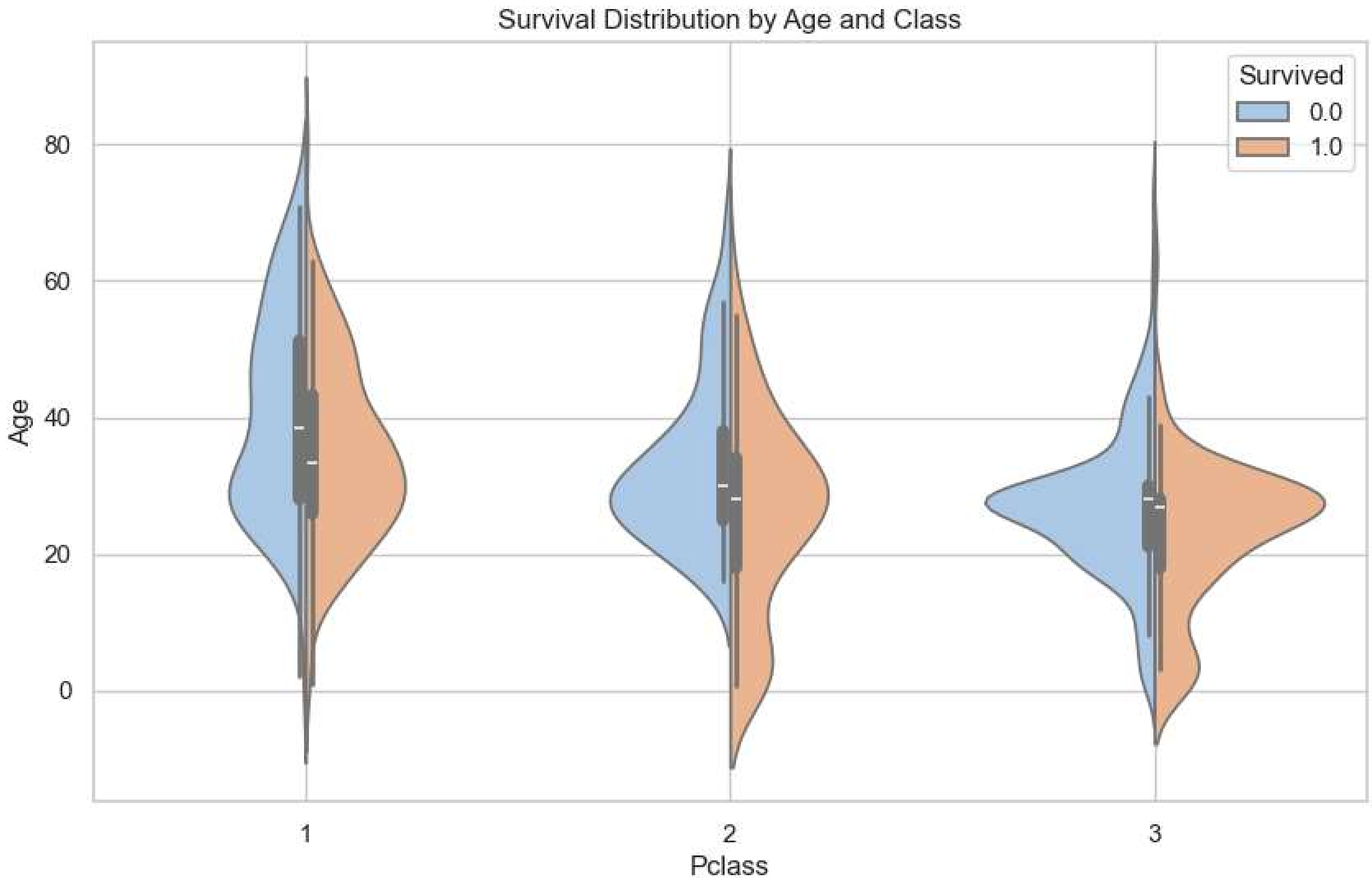
```
# Create pivot table of survival rates
pivot = train.pivot_table(index='Pclass', columns='Sex', values='Survived')

# Plot as heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(pivot, annot=True, cmap='YlGnBu', fmt='.2f')
plt.title('Survival Rate by Passenger Class and Gender')
plt.show()
```



📌 Observations: - Females in all classes had higher survival rates than males. - 1st class females had the best survival outcomes.

🧪 Violinplot to Show Distribution + Survival

In [88]:
```
plt.figure(figsize=(10, 6))
sns.violinplot(data=train, x='Pclass', y='Age', hue='Survived', split=True, palette='pastel')
plt.title('Survival Distribution by Age and Class')
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.grid(True)
plt.show()
```



📌 Observations: - Many young survivors in 1st and 2nd class. - Older people had lower survival across classes.

🛠️ Feature Engineering (Creating Powerful New Features)

🔹 Family Size = SibSp + Parch + You

In [92]:
```
# Total number of family members onboard (self included)
combined['FamilySize'] = combined['SibSp'] + combined['Parch'] + 1
```

◆ IsAlone = 1 if FamilySize == 1

In [95]:
```
# Flag if passenger is traveling alone
combined['IsAlone'] = 0
combined.loc[combined['FamilySize'] == 1, 'IsAlone'] = 1
```

◆ Extract Title from Name

In [98]:
```
# Extract title using regex from Name
combined['Title'] = combined['Name'].str.extract(' ([A-Za-z]+)\.', expand=False)
```

In [100…
```
# Combine rare titles under 'Rare'
combined['Title'] = combined['Title'].replace(['Lady', 'Countess','Capt', 'Col', 'Don', 'Dr',
        'Major', 'Rev', 'Sir', 'Jonkheer', 'Dona'], 'Rare')

# Standardize similar titles
combined['Title'] = combined['Title'].replace(['Mlle', 'Ms'], 'Miss')
combined['Title'] = combined['Title'].replace('Mme', 'Mrs')
```

In [102…
```
combined['Title'].value_counts()
```

Out[102…
```
Title
Mr        757
Miss      264
Mrs       198
Master     61
Rare       29
Name: count, dtype: int64
```

◆ Optional: Age Bins (e.g. Child, Teen, Adult, Senior)

In [105…
```
combined['AgeBin'] = pd.cut(combined['Age'], bins=[0, 12, 18, 35, 60, 80],
                            labels=['Child', 'Teen', 'YoungAdult', 'Adult', 'Senior'])
```

◆ Optional: Fare Bins

In [108…
```
combined['FareBin'] = pd.qcut(combined['Fare'], 4, labels=['Low', 'Medium', 'High', 'Very_High'])
```

In [110…
```
combined[['FamilySize', 'IsAlone', 'Title', 'AgeBin', 'FareBin']].head()
```

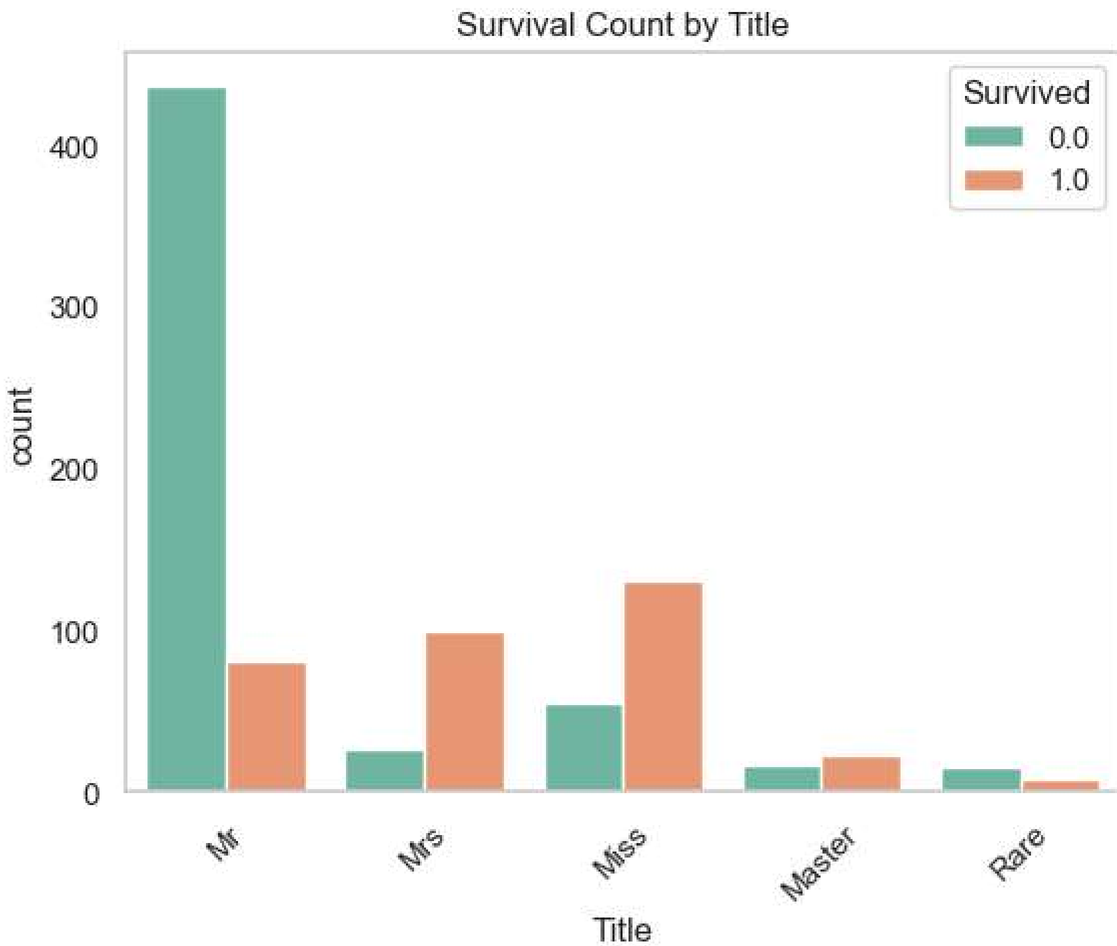
Out[110…

	FamilySize	IsAlone	Title	AgeBin	FareBin
0	2	0	Mr	YoungAdult	Low
1	2	0	Mrs	Adult	Very_High
2	1	1	Miss	YoungAdult	Medium
3	2	0	Mrs	YoungAdult	Very_High
4	1	1	Mr	YoungAdult	Medium

Visualize

Survival Rate by Title

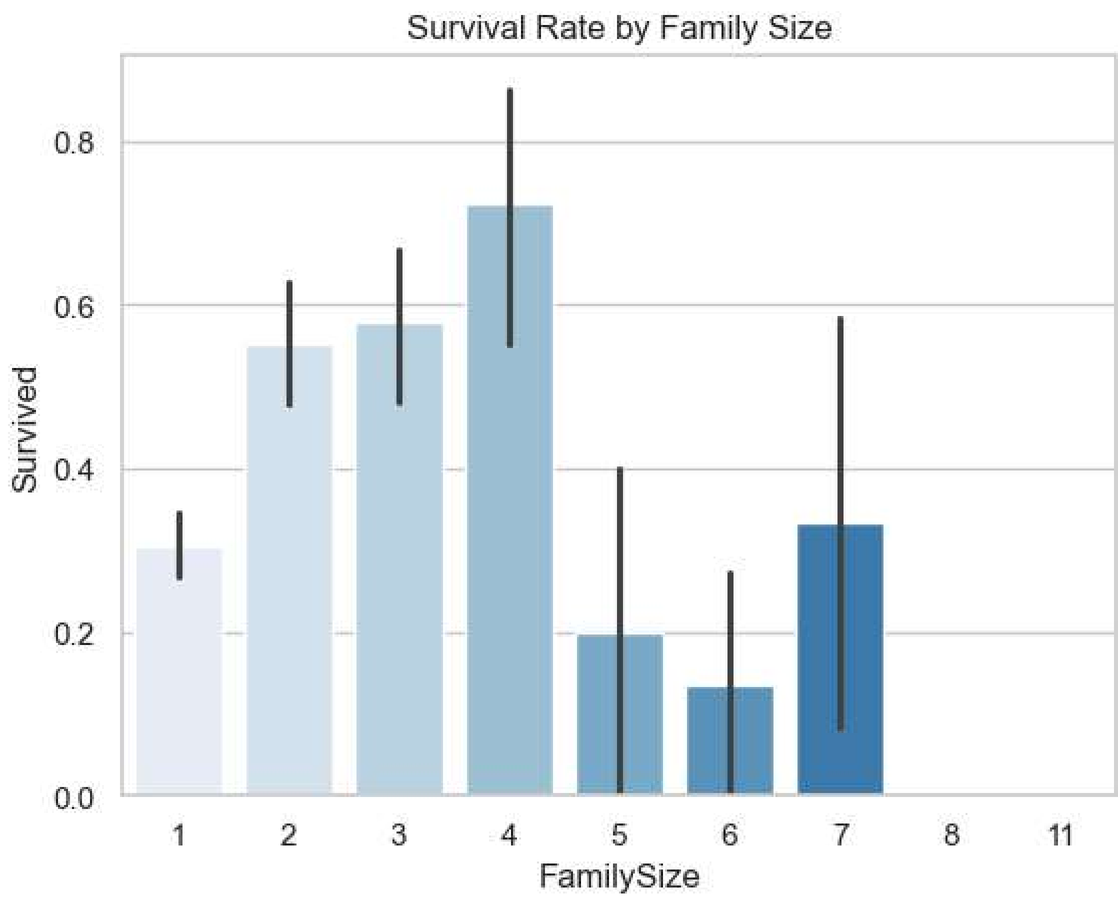
In [114…
```
sns.countplot(data=combined[combined['source']=='train'], x='Title', hue='Survived', palette='Set2')
plt.title('Survival Count by Title')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()
```



Survival by Family Size

In [117…

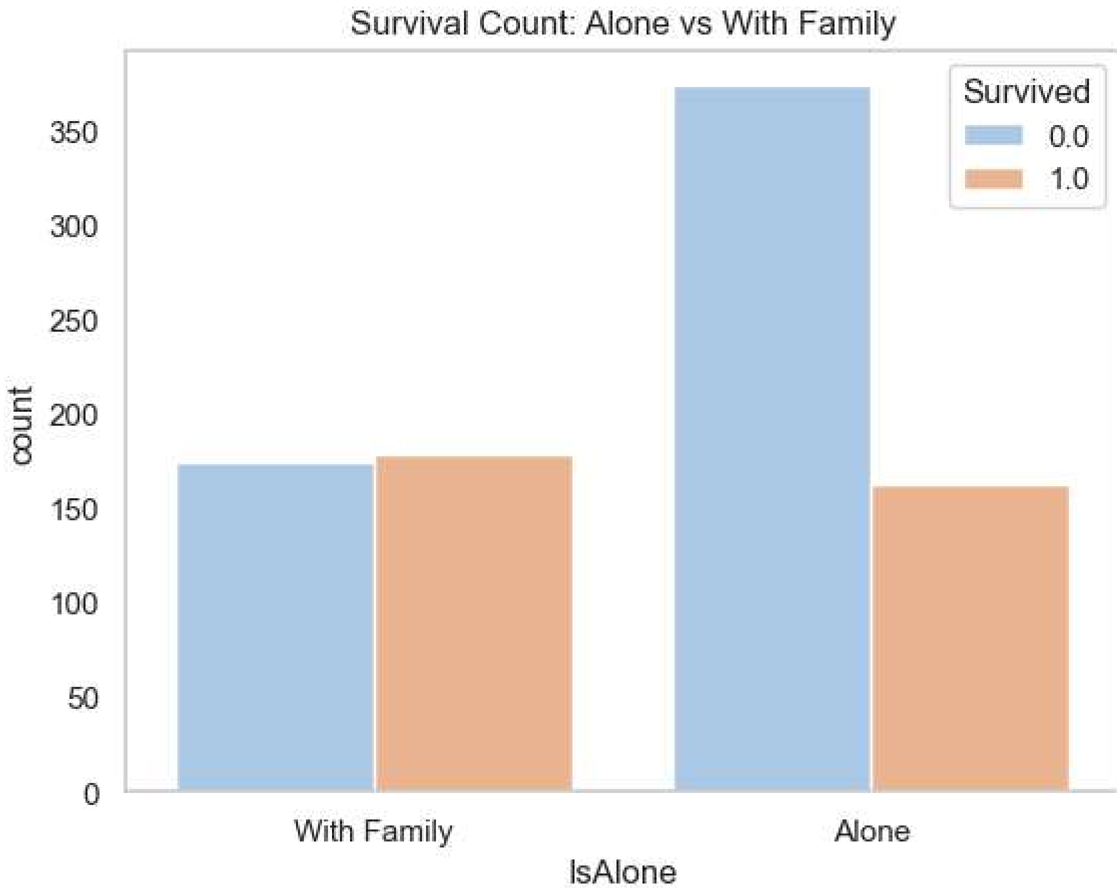
```
sns.barplot(data=combined[combined['source']=='train'], x='FamilySize', y='Survived', palette='Blues')
plt.title('Survival Rate by Family Size')
plt.show()
```



IsAlone vs Survival

In [120…

```
sns.countplot(data=combined[combined['source']=='train'], x='IsAlone', hue='Survived', palette='pastel')
plt.title('Survival Count: Alone vs With Family')
plt.xticks([0, 1], ['With Family', 'Alone'])
plt.grid(axis='y')
plt.show()
```



✅ Summary of Findings + Optional Prediction

📌 Summary of Key Findings

- 🎯 **Sex vs Survival**: Females had a significantly higher survival rate than males.
- 💰 **Fare vs Survival**: Passengers who paid higher fares were more likely to survive.
- 🏷️ **Pclass vs Survival**: 1st class passengers had the best survival rate; 3rd class the worst.
- 👨‍👩‍👧 **FamilySize**: Those traveling alone had lower survival rates.
- 👶 **Age**: Children had a higher survival rate; middle-aged adults saw more casualties.
- 🧱 **Deck**: Decks B, D, and E had higher survival rates; these often housed 1st class passengers.

- 🚢 **Embarked**: Port 'C' passengers had higher survival rates than those from 'S' or 'Q'.
- 🧠 **Title (from Name)**: Social titles revealed insights — "Miss" and "Mrs" survived more than "Mr".

✅ Feature engineering (FamilySize, IsAlone, Title, AgeBin) added valuable patterns to enhance modeling or story-building.

Simple Rule-Based Prediction

In [ ]:
```
# Recreate test set from combined
test_data = combined[combined['source'] == 'test'].copy()

# Simple rule: female = survive, male = not survive
test_data['Survived'] = 0  # default: not survived
test_data.loc[test_data['Sex'] == 'female', 'Survived'] = 1
```

📤 Create Submission File

In [ ]:
```
# Save predictions in submission format
submission = test_data[['PassengerId', 'Survived']]
submission.to_csv('simple_gender_submission.csv', index=False)
```

In [ ]: